

SDN Anomaly Detection Using Isolation Forest

Paper Review and Practical Demonstration

Computer Networking

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Why Anomaly Detection Matters



Modern networks are larger

Scale and heterogeneity (cloud, IoT, data centers) increase complexity.



Massive traffic volumes

High throughput and ephemeral flows hide anomalies in noise.



Manual detection is infeasible

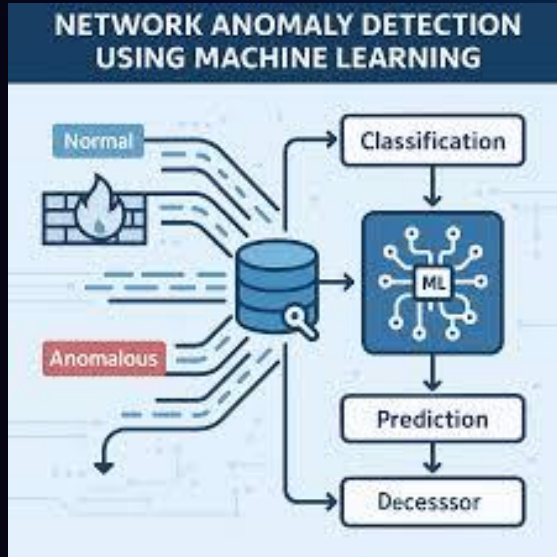
Automated detection required for timely response.



Early detection improves security

Reduce dwell time, limit damage, enable faster mitigation.

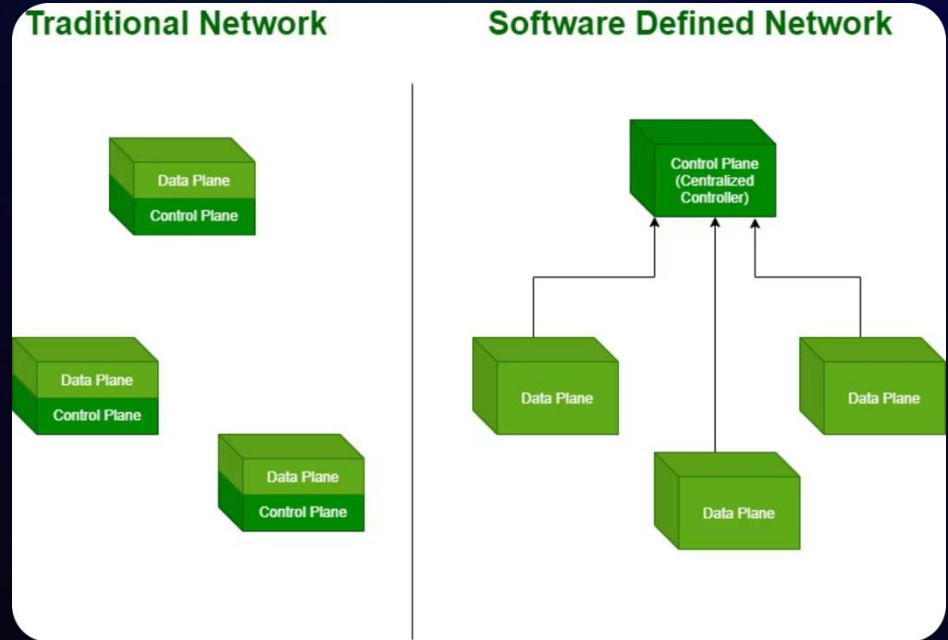
Anomaly in Network Traffic



- Behavior outside normal traffic distribution
- Indicator of attack, failure, or misconfiguration
- Examples: sudden spike, unusual volume, elevated error rates
- Goal: detect early, minimize false alarms

Software-Defined Networking

- Control plane separated from packet forwarding
- Centralized controller with global network view
- Switches handle fast forwarding
- Programmable policies and telemetry



SDN as a Security Enabler



Centralized visibility

Controller sees aggregated flow-level telemetry across the fabric.



Faster response

Controller can push policies to mitigate threats in near real-time.



Flexible enforcement

Dynamic actions: alert, monitor, rate-limit, or block flows.

flow: Traffic → Switch → Controller → Security Action

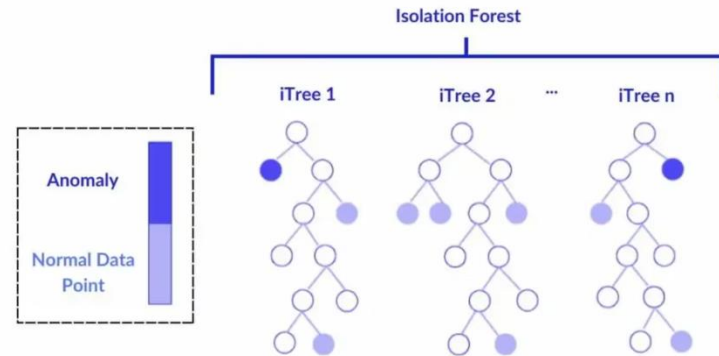
How Isolation Forest Detects Anomalies

Anomalies are few and inherently different

Random partitioning isolates samples into shorter paths

Short path length $\Rightarrow$ higher anomaly score

Model produces an anomaly score per flow or sample



Selected Paper: Main Idea



Evaluate Isolation Forest in SDN

Investigates feasibility of isolation-based detection on SDN telemetry.



Feature-driven approach

Extracts flow-level features relevant to anomalies.



SDN response loop

Links detection output to controller actions for mitigation.

J. Electrical Systems 19-4 (2023): 279-297

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Evaluating the Isolation Forest Method for Anomaly Detection in Software-Defined Networking Security



Abstract: - The research addresses the critical anomaly detection problem in Software-Defined Networking (SDN), a domain where network integrity and security are paramount. Employing the Isolation Forest algorithm, a machine learning model renowned for its efficacy in identifying outliers, the study systematically generates synthetic network traffic data to train and test the model's detection capabilities. The methodology encompasses simulating a range of contamination rates to reflect varying degrees of anomalous activities within the network. Key findings indicate that while the model exhibits potential in anomaly detection, as reflected by the progressive increase in triggered alerts and policy changes, its performance metrics, such as precision, recall, F1-score, and AUC, reveal limitations in its current application. The research contributes to the field by providing a detailed analysis of the Isolation Forest algorithm's performance in an SDN context and laying the groundwork for future enhancements in machine learning-based security measures within these networks.

Keywords: Anomaly Detection, Software-Defined Networking, Isolation Forest, Synthetic Data Simulation, Network Security.

Paper Methodology

01

Synthetic traffic generation

Create benign and anomalous flow patterns to test detection.

03

Isolation Forest model

Train on features, compute anomaly scores per sample.

02

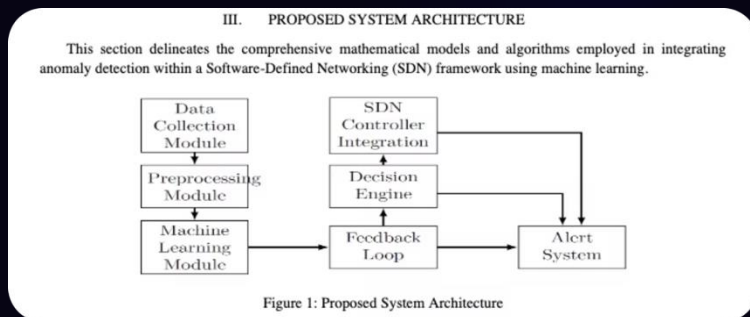
Feature extraction & preprocessing

Flow features (rates, bytes, durations) and normalization.

04

Decision & evaluation

Thresholding, mapping scores to SDN actions, assess quality.



Paper Takeaways

Key points

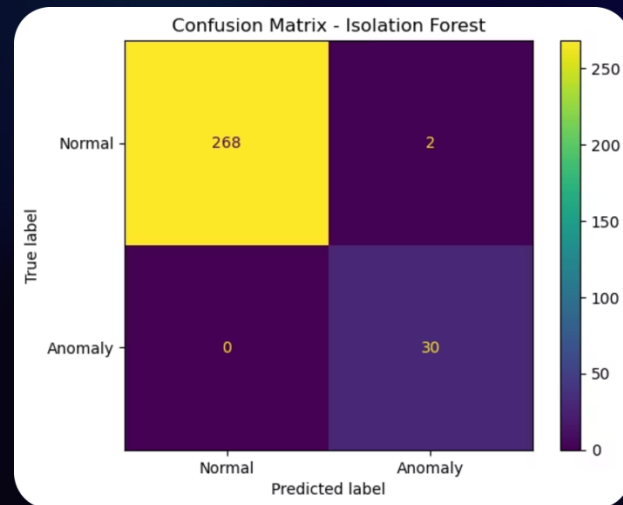
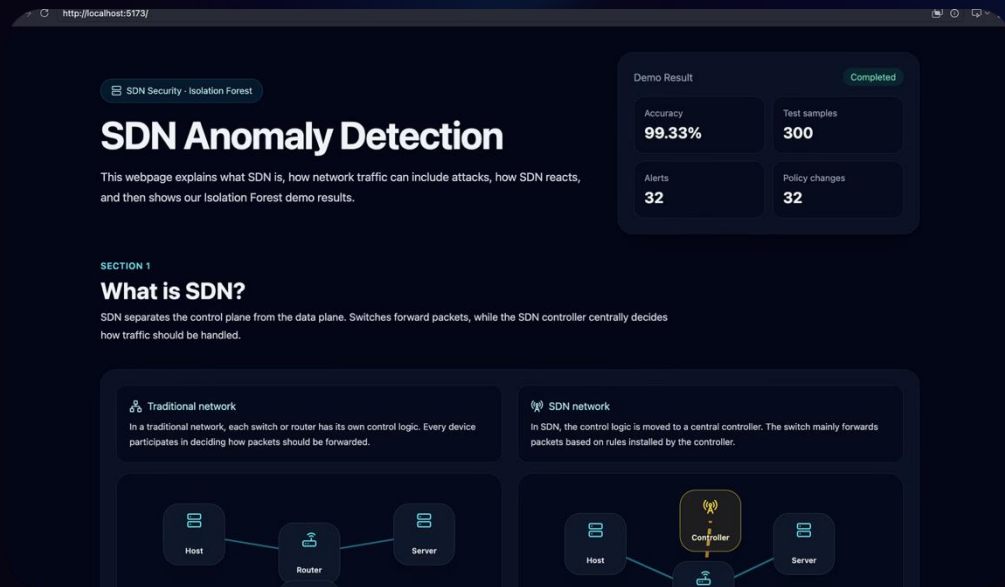
- Feature choice strongly affects detection
- Synthetic data validates concept quickly
- Isolation Forest is lightweight and unsupervised
- Controller integration enables automated mitigation

Limitations

- Synthetic traffic may not reflect real variability
- Evaluation metrics show room for improvement
- Thresholds may not generalize across networks
- Real-world SDN deployment complexities remain

Practical Implementation — Live Demo

Note: we will switch to the live Python model and React demo now. Slides intentionally exclude implementation results and summary of our model . Available on [Github](#)



Thank you for your consideration.

We appreciate your time and attention today